Activity

- A researcher wants to study the relationship between **hours studied (x)** and **exam score (y)** for a group of students. The data collected is:

x (Hours)	y (Score)
1	50
2	55
3	65
4	70
5	75

Tasks

1. Find the **regression coefficients**:
    1. Intercept b_0
    2. Slope b_1
2. Write the **regression equation**.
3. Compute the following:
    1. Total Sum of Squares (**SST**)
    2. Regression Sum of Squares (**SSR**)
    3. Error Sum of Squares (**SSE**)
4. Calculate:
    1. Coefficient of determination (**r^2**)
    2. Correlation coefficient (**r_{xy}**)
5. Use the regression model to **predict the exam score** for a student who studies **6 hours**.